

## SAFETY DATA SHEET

Creation Date 24-November-2010

Revision Date 25-March-2024

Revision Number 3

### 1. Identification

**Product Name** Bromine liquid

**Cat No. :** R14300

**CAS-No** 7726-95-6  
**Synonyms** Bromine molecule.; Diatomic bromine; Dibromine

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                          |              |
|------------------------------------------|--------------|
| <b>Acute Inhalation Toxicity</b>         | Category 1   |
| <b>Skin Corrosion/Irritation</b>         | Category 1 A |
| <b>Serious Eye Damage/Eye Irritation</b> | Category 1   |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Fatal if inhaled

Causes severe skin burns and eye damage

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Wear respiratory protection

**Response**

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Wash contaminated clothing before reuse

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic organisms

### 3. Composition/Information on Ingredients

| Component | CAS-No    | Weight % |
|-----------|-----------|----------|
| Bromine   | 7726-95-6 | >95      |

### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Seek immediate medical attention/advice.                                                                                                                                                                                                            |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Seek immediate medical attention/advice.                                                                                                                                                                                                                                  |
| <b>Inhalation</b>                      | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician or poison control center immediately. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                               |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation                                              |

## Notes to Physician

Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

**Unsuitable Extinguishing Media** No information available

**Flash Point** Not applicable  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

Very toxic by inhalation. May be fatal if inhaled. Corrosive material. May intensify fire; oxidizer. Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Hydrogen halides. Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA**

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
OX

## 6. Accidental release measures

**Personal Precautions** Wear self-contained breathing apparatus and protective suit. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing.

**Environmental Precautions** Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up** Wear self-contained breathing apparatus and protective suit. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Incompatible Materials. Organic materials. Strong oxidizing agents. Ammonia. Fluorine. Metals. Reducing Agent.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component | Alberta      | British Columbia | Ontario TWAEV | Quebec       | ACGIH TLV    | OSHA PEL       | NIOSH       |
|-----------|--------------|------------------|---------------|--------------|--------------|----------------|-------------|
| Bromine   | TWA: 0.1 ppm | TWA: 0.1 ppm     | TWA: 0.1 ppm  | TWA: 0.1 ppm | TWA: 0.1 ppm | (Vacated) TWA: | IDLH: 3 ppm |

|  |                                                                            |               |               |                                                                             |               |                                                                                                                                                                 |                                                                                          |
|--|----------------------------------------------------------------------------|---------------|---------------|-----------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|  | TWA: 0.7 mg/m <sup>3</sup><br>STEL: 0.2 ppm<br>STEL: 1.3 mg/m <sup>3</sup> | STEL: 0.2 ppm | STEL: 0.2 ppm | TWA: 0.66 mg/m <sup>3</sup><br>STEL: 0.2 ppm<br>STEL: 1.3 mg/m <sup>3</sup> | STEL: 0.2 ppm | 0.1 ppm<br>(Vacated) TWA: 0.7 mg/m <sup>3</sup><br>(Vacated) STEL: 0.3 ppm<br>(Vacated) STEL: 2 mg/m <sup>3</sup><br>TWA: 0.1 ppm<br>TWA: 0.7 mg/m <sup>3</sup> | TWA: 0.1 ppm<br>TWA: 0.7 mg/m <sup>3</sup><br>STEL: 0.3 ppm<br>STEL: 2 mg/m <sup>3</sup> |
|--|----------------------------------------------------------------------------|---------------|---------------|-----------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Butyl rubber   | See manufacturers | -               | Splash protection only |
| Natural rubber | recommendations   |                 |                        |
| Nitrile rubber |                   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Acid gases filter Type E Yellow Inorganic gases and vapours filter Type B Grey

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

**Physical State**

Liquid

**Appearance**

Red brown

**Odor**

Strong

**Odor Threshold**

No information available

|                                        |                          |
|----------------------------------------|--------------------------|
| pH                                     | No information available |
| Melting Point/Range                    | -7.2 °C / 19 °F          |
| Boiling Point/Range                    | 58.7 °C / 137.7 °F       |
| Flash Point                            | Not applicable           |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | 230 mbar @ 20 °C         |
| Vapor Density                          | 5.51 (Air = 1.0)         |
| Specific Gravity                       | 3.111                    |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | 0.314 cs at 25 °C        |
| Molecular Formula                      | Br <sub>2</sub>          |
| Molecular Weight                       | 159.82                   |

## 10. Stability and reactivity

|                                  |                                                                                            |
|----------------------------------|--------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                 |
| Stability                        | Stable under normal conditions. May intensify fire; oxidizer.                              |
| Conditions to Avoid              | Incompatible products. Excess heat.                                                        |
| Incompatible Materials           | Organic materials, Strong oxidizing agents, Ammonia, Fluorine, Metals, Reducing Agent      |
| Hazardous Decomposition Products | Hydrogen halides, Thermal decomposition can lead to release of irritating gases and vapors |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                   |
| Hazardous Reactions              | None under normal processing.                                                              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component | LD50 Oral                 | LD50 Dermal | LC50 Inhalation             |
|-----------|---------------------------|-------------|-----------------------------|
| Bromine   | LD50 = 2600 mg/kg ( Rat ) | Not listed  | LC50 = 2.7 mg/L (Rat, 4hrs) |

Toxicologically Synergistic Products No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes severe burns by all exposure routes                                               |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|-----------|------------|------------|------------|------------|------------|
| Bromine   | 7726-95-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

Mutagenic Effects No information available

Reproductive Effects No information available.

|                                                   |                                                                                                                                                                                                                                                            |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Developmental Effects</b>                      | No information available.                                                                                                                                                                                                                                  |
| <b>Teratogenicity</b>                             | No information available.                                                                                                                                                                                                                                  |
| <b>STOT - single exposure</b>                     | None known                                                                                                                                                                                                                                                 |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                                                                                                                                                 |
| <b>Aspiration hazard</b>                          | No information available                                                                                                                                                                                                                                   |
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                                                                                                                                                   |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                             |

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

|                                      |                                                                 |
|--------------------------------------|-----------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely based on information available.         |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                       |
| <b>Mobility</b>                      | Will likely be mobile in the environment due to its volatility. |

| Component | log Pow |
|-----------|---------|
| Bromine   | 1.03    |

## 13. Disposal considerations

|                               |                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste Disposal Methods</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 14. Transport information

### DOT

|                                |         |
|--------------------------------|---------|
| <b>UN-No</b>                   | UN1744  |
| <b>Proper Shipping Name</b>    | BROMINE |
| <b>Hazard Class</b>            | 8       |
| <b>Subsidiary Hazard Class</b> | 6.1     |
| <b>Packing Group</b>           | I       |

### TDG

|                                |         |
|--------------------------------|---------|
| <b>UN-No</b>                   | UN1744  |
| <b>Proper Shipping Name</b>    | BROMINE |
| <b>Hazard Class</b>            | 8       |
| <b>Subsidiary Hazard Class</b> | 6.1     |
| <b>Packing Group</b>           | I       |

### IATA

|                                |                                      |
|--------------------------------|--------------------------------------|
| <b>UN-No</b>                   | UN1744                               |
| <b>Proper Shipping Name</b>    | BROMINE FORBIDDEN FOR IATA TRANSPORT |
| <b>Hazard Class</b>            | 8                                    |
| <b>Subsidiary Hazard Class</b> | 6.1                                  |
| <b>Packing Group</b>           | I                                    |

### IMDG/IMO

|                                |         |
|--------------------------------|---------|
| <b>UN-No</b>                   | UN1744  |
| <b>Proper Shipping Name</b>    | BROMINE |
| <b>Hazard Class</b>            | 8       |
| <b>Subsidiary Hazard Class</b> | 6.1     |

## Packing Group

I

## 15. Regulatory information

## International Inventories

| Component | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Bromine   | 7726-95-6 | X   | -    | X    | ACTIVE                                        | 231-778-1 | -      | -   |

| Component | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|-----------|-------|----------|------|------|------|------|-------|-------|
| Bromine   | 7726-95-6 | X     | KE-03605 | X    | -    | X    | X    | X     | X     |

## Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Bromine   | Part 1, Group A Substance                            |                                                                            |                                           |

## Other International Regulations

## Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Bromine   | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Bromine   | 7726-95-6 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Bromine   | 7726-95-6 | 20 tonne                                                                                  | 100 tonne                                                                                | Not applicable             | Not applicable                     |

**16. Other information**

|                         |                                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------|
| <b>Prepared By</b>      | Product Safety Department<br>Email: chem.techinfo@thermofisher.com<br>www.thermofisher.com |
| <b>Creation Date</b>    | 24-November-2010                                                                           |
| <b>Revision Date</b>    | 25-March-2024                                                                              |
| <b>Print Date</b>       | 25-March-2024                                                                              |
| <b>Revision Summary</b> | New emergency telephone response service provider.                                         |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**